

Veil RFM Analytics

Retail customer analytics platform — RFM segmentation, Markov Chain transition modeling, and AI-powered chatbot. 100% TypeScript, runs on Cloudflare Workers + Pages + Qwen LLM.

Forked and modernized from the original [veil-dotnet](#) R/Python project.

Architecture

```
veil-rfm/
├─ packages/
│  └─ core/           # Pure TS: RFM engine + Markov Chain + What-If logic
│  └─ worker/         # Cloudflare Worker (REST API + Qwen chatbot proxy)
│  └─ frontend/       # Cloudflare Pages (React dashboard)
├─ scripts/           # Test scripts (250 Q&A validation)
├─ logs/              # Chatbot Q&A test logs
└─ sample-data/      # Sample CSV for testing
```

Package: core

Pure business logic, zero I/O, fully unit-tested (vitest).

Module	Source	Purpose
rfm.ts	findRFM.R port	RFM scoring (1-5) + 11-segment classification
markov.ts	rfmMC.R port	Markov Chain transition matrix + prediction
clumpiness.ts	clumpiness.R port	Inter-event time clumpiness metric
whatif.ts	New	What-If simulation engine

Package: worker

Cloudflare Worker with 5 REST endpoints:

Endpoint	Method	Description
/api/rfm	POST	Compute RFM scores and segments
/api/rfm/transition	POST	Markov Chain transition probability matrix
/api/rfm/predict	POST	Predict future segment distribution
/api/rfm/whatif	POST	Simulate behavioral changes
/api/chat	POST	Qwen-powered chatbot with 14 function calls
/api/health	GET	Health check

14 Chatbot Function Calls: `getCustomerInfo`, `listAllCustomers`, `runWhatIf`, `explainSegment`, `getSegmentDistribution`, `suggestTargetSegment`, `getSegmentStats`, `getAtRiskCustomers`, `compareCustomers`, `getCustomersByFilter`, `getRevenueBySegment`, `getNewVsReturning`, `getSummaryStats`, `getSegmentMigration`

Package: frontend

React 18 + TypeScript dashboard with 3-language i18n (EN / 繁體中文 / 简体中文).

Tab structure (mirrors original R Shiny layout):

- Overview — Pie chart + segment distribution table
- Characteristics — R/F/M bar charts per segment, frequency histogram
- Segment Transition — Interactive vis-network graph + prediction line chart
- Customer Summary — Filterable data table with CSV export
- What-If Analysis — Slider-based simulation with target segment suggestions
- LTV Overview — Average lifetime value per segment bar chart

Right panel: Resizable Qwen-powered chatbot with multi-turn conversation.

Quick Start

Prerequisites

- Node.js 20+
- Cloudflare account (for deployment)
- Qwen API key from [DashScope](#) (for chatbot)

Local Development

```
# Install dependencies
cd veil-rfm
npm install

# Run core unit tests (13 tests)
npm test

# Start Worker API (localhost:8787)
npm run dev:worker

# Start frontend (localhost:3000)
npm run dev:frontend
```

Deploy to Cloudflare

```
# Set Qwen API key (for chatbot)
npx -w packages/worker wrangler secret put QWEN_API_KEY

# Deploy Worker API
npm run deploy:worker

# Build and deploy frontend
VITE_API_URL=https://your-worker.workers.dev npm run build -w packages/frontend
npx -w packages/frontend wrangler pages deploy dist --project-name veil-rfm
```

Custom Domain

```
# Add custom domain to Pages project
npx wrangler pages domain add veil-rfm your-domain.com
```

Add a CNAME DNS record pointing to `veil-rfm.pages.dev` (proxied).

Usage Guide

Uploading Data

- 1. Open the dashboard
- 2. Navigate to **Import** in the sidebar
- 3. Drag-and-drop a CSV file with these columns:

```
MemberID, OrderID, Timestamp, NetPrice, Quantity, ProductID, ProductName, Category
```

- 4. Or click **Load Sample Data** for a demo with 8 customers

Understanding RFM Segments

#	Segment	RFM Pattern	Business Action
1	Best Customers	555	VIP retention, exclusive previews
2	Loyal Customers	≥ 444	Loyalty rewards, cross-sell
3	Potential Loyalist	≥ 333	Nurture campaigns, milestone rewards
4	Low-spending Active Loyal	$R \geq 4, F \geq 4, M \leq 2$	Bundle deals, upsell
5	High-spending New Customers	$R \geq 4, M \geq 4, F \leq 2$	Follow-up sequence, 2nd purchase discount
6	Almost Lost Customers	$R = 2-3, F \geq 4, M \geq 4$	Win-back campaign NOW
7	Churned Best Customers	$R = 1, F \geq 4, M \geq 4$	Aggressive win-back, survey
8	Customers Needing Attention	Mixed	Investigate by category/channel
9	About to Sleep Customers	≤ 333	Re-engagement campaign
10	Hibernating Customers	≤ 222	Low-cost reactivation test
11	Lost Cheap Customers	111	Minimal marketing spend

Using the Chatbot

The right-side chatbot is powered by Qwen LLM with 14 function calls. Example questions:

- "Who are the top 5 customers by spending?"
- "Which customers are at risk of churning?"
- "What if C003 made 5 more purchases?"
- "Compare C001 and C004"
- "How much revenue does each segment generate?"
- "What changes would move C008 to Potential Loyalist?"
- 解釋 Best Customers 是什麼？
- 顯示所有客戶

Running Q&A Validation

```
bash scripts/test-chatbot.sh
```

Tests 250 questions across 14 categories. Results saved to `logs/` .

Tech Stack

Layer	Technology
Frontend	React 18, TypeScript, Vite, Tailwind CSS, Recharts, vis-network
API	Cloudflare Workers, TypeScript
AI Chatbot	Qwen-plus via DashScope (OpenAI-compatible)
Testing	Vitest
Deployment	Wrangler (Workers + Pages)
i18n	Custom React context (en / zh-TW / zh-CN)

Project Structure

```
packages/core/src/
├─ index.ts           # Barrel export
├─ types.ts           # TypeScript interfaces + 11 segment constants
├─ rfm.ts             # RFM scoring engine
├─ markov.ts          # Markov Chain transition engine
├─ clumpiness.ts      # Clumpiness metric
└─ whatif.ts         # What-If simulation engine

packages/worker/src/
├─ index.ts           # Worker entry + REST endpoints
└─ chat.ts            # Qwen chatbot proxy + 14 function implementations

packages/frontend/src/
├─ main.tsx           # Entry point + i18n provider
├─ App.tsx            # Shell: header, sidebar, KPI bar, routes
├─ index.css          # Enterprise design system
├─ lib/
│   ├─ api.ts         # API client
│   ├─ i18n.ts        # 3-language translations
│   ├─ parse.ts       # CSV parser (Papa Parse)
│   └─ sampleData.ts  # Embedded demo transactions
├─ components/
│   ├─ ChatBot.tsx    # Qwen chatbot panel (resizable)
│   ├─ FileUpload.tsx # CSV drag-and-drop
│   └─ DashboardSections.tsx
└─ pages/
    ├─ RFMOverview.tsx
    ├─ RFMCharacteristics.tsx
    ├─ RFMTransition.tsx
    └─ RFMCustomerSummary.tsx
```

└─ LTV0verview.tsx
└─ WhatIf.tsx

Verified Q&A Coverage

250 questions tested across 14 categories with 100% pass rate (see [logs/](#)):

Category	Questions	Status
Customer Lookup	20	100%
Segment Explanation	15	100%
Segment Distribution	10	100%
Rankings	20	100%
What-If Simulation	25	100%
Churn & Risk	20	100%
Revenue Analysis	20	100%
Customer Comparison	15	100%
Filtering & Search	20	100%
New vs Returning	10	100%
Summary & KPIs	15	100%
Migration & Transition	15	100%
Business Actions	25	100%
Edge Cases	20	100%
Total	250	100%